Lecture 2: Introduction to Linux Notes

1. **What is an Operating System?** An Operating System offers all software related features of a computer and provides functionality for the user when operating the computer's hardware.
2. **What is a kernel?** A Kernel is a component within software that manages low-level features of the computer like memory allocation and interaction between programs.
3. **Which other parts aside from the kernel identify an OS?** Other components of an Operating System that so not include a kernel are the Command-Line Shells, Libraries, and Graphical User Interfaces.
4. **What is Linux?** Linux is a Unix-like Operating System that consists of a kernel, libraries, and utilities. Linux is known for being highly scalable and free of charge.
5. **What is a Linux distribution?** A Linux distribution is the complete linux system package that contain the kernel, core unix tools, supplemental software, startup scripts, and an installer. The full distribution is simply collections of many files in a package.
6. **List at least 4 Linux characteristics:**
    1. Linux is open source software
    2. Linux is available free of charge
    3. YOu can install LInux on almost any system
    4. Many business rely on Linux for day to day operations
    5. Linux includes multiple server programs and programming languages upon download.
7. **What is Debian?** Debian is a volunteer-based organization that promotes the ideals of the Free Software Community in addition to developing free software.
8. **List and define the different types of licensing agreements**
    - **Open Source Licensing** is the agreement that software may be distributed for a fee or free. The source code is available alongside the software.
    - **Closed Source Licensing** is the agreement that software is not distributed with the source code, restricting users from modifying the code
        - **Freeware** provides software at no charge but the source code is unavailable
        - **Shareware** provides the software on a free-trial basis.
        - **Free Software** is the agreement that software is distributed with the source code and can be free or obtained by a fee.
9. **What is Free Software? Define the 4 freedoms.** Free Software is the agreement that software is distributed with the source code but can be obtained for a fee. The freedoms defined by the Free Software Foundation (FSF) include:
    1. Use the software for any purpose
    2. examine the source code and modify it as you see fit
    3. redistribute the software
    4. redistribute modified software

10. **What is virtualization?** Virtualization is is the use of multiple Operating Systems on one physical machine. Virtualization allows for a division of hardware and the creation of multiple computers inside one machine offering, cheaper hardware.